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SW Engineering CSC 648/848 [02]

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Use Cases, High Level Requirements and Architecture

Milestone 1

Project Name: LifeTrack

Team 01

1. Executive Summary

As college students, we are often trying to balance our academic, professional, and personal lives. There is an unprecedented challenge that they face: managing academic schedules, building professional careers, maintaining wellness, and developing life skills. All aspects are being juggled simultaneously and often without adequate tools or support. Solutions are spread across multiple apps and platforms, forcing students to switch back and forth from separate systems for goal tracking, resource discovery, calendar management, and reflection. This inconvenient fragmentation leads to decreased productivity, missed opportunities, and increased stress as students struggle to maintain visibility across all areas of their lives. The market lacks a unified, student-centered platform that integrates goal management, resource discovery, and progress analytics into a single cohesive system designed specifically for the college experience. Our project, LifeTrack, addressed this critical gap by providing the first comprehensive, highly personalized life management platform built from the ground up to meet the unique needs of modern college students.

LifeTrack is a web-based platform that aids college students in organizing tasks and achieving their academic, professional, and personal goals through a unified dashboard and powerful resource discovery system. The platform's core features include a personalized student dashboard where users manage short and long-term goals, organize multimilestone projects, and track progress through visual analytics and end-of-semester growth reports. The system tracks progress through visual analytics, generates end-of-semester personal growth reports, and provides insights on academic workload, stress indicators, and skill development patterns. An advanced search and discovery engine enables students to explore thousands of categorized resources, including study tools, career development materials, wellness resources, and filtered

by use case, skill area, and personal goals. Students can save resources directly to their dashboard, link them to specific goals and projects, and organize them into personal collections for easy access. A unique reflection feature encourages students to document their learning journey through structured weekly progress challenges and learning insights, with all reflections linked to specific goals and milestones for long-term growth tracking. The platform also fosters community learning through public sharing of successful goal templates, workflows, and resources, with a rating and bookmarking feature that helps students discover proven strategies from their peers. LifeTrack is uniquely designed to be a purpose-built design for the college student experience, which is what sets it apart from other generic productivity apps or academic planning tools. This new application will help students make informed decisions about where to invest their time and energy.

Team 01 is a six-member student development team from San Francisco State University's Software Engineering program, combining technical expertise in full-stack web development, database design, and user experience with a firsthand understanding of the challenges facing today's college students. LifeTrack is planned to be started by Khaterina, Mohit, John, Will, Alex, and Lasiru. We are committed to providing the best personalized platform for college students to thrive in all aspects of their lives. Team 01 believes this application is what is needed for college students and will transform life management into a much easier practice.

2. Personas and Main Use Cases:

1. Personas:

- a. College students
- b. Faculty and Academic Programs

- c. University Support Organizations

2. Main Use Cases:

- a. Studying
- b. Skill Building
- c. Job Preparation
- d. Mental Wellness

College Students:

General Characteristics: A student can vary from first-year undergraduates, transfer students, and graduate students. They may generally have one or multiple responsibilities such as managing several courses, preparing for interviews or internships, currently working, and maintaining a healthy social life. Most of these students are comfortable using technology.

Goals: Students aim to succeed academically, stay organized, develop skills for future employment, and maintain a healthy balance of their mental wellness and coursework.

Skills: The majority of students have a high proficiency in using different digital tools to assist in coursework and research. They are attuned with current events both politically and socially.

Pain Points: One of the biggest struggles for students is time management and lack experience utilizing planning tools. They may also struggle with seeking resources for mental wellness, and job preparation.

Faculty and Academic Programs:

General Characteristics: Faculty members and academic programs may be responsible for supporting students' academic success, teaching, research, maintaining campus infrastructure, general management, and facilitating resources for students.

Goals: The main goals for faculty and certain academic programs is to improve student engagement, facilitate skill development, and ensure that students meet the required academic targets.

Skills: Faculty members are professionally trained and accredited in their specialization. They are proficient at using different digital tools and deploying them to help students' success.

Pain Points: Faculty members may have difficulty monitoring student progress outside of the classroom and identifying students who are struggling. They may face challenges incorporating wellness support and career readiness into the course. Another issue may be limited access to resources they are able to obtain and provide.

University Support Organizations

General Characteristics: University Support Organizations can be made up of career centers, tutoring programs, counseling services, and mental wellness programs. Each program aims to help students succeed by providing the necessary resources and environments. These programs are not limited to academia.

Goals: These organizations aim to help with student retention, promote mental wellness, and give students all the skills needed to succeed beyond college.

Skills: The staff in the organizations are professionally trained or accredited, have administrative experience, and communication expertise.

Pain Points: The organizations may have difficulty tracking student success in their individual programs, delays in providing timely support, budget obstacles, and limited access to resources outside of the college itself.

Use Case 1: Studying

Actor: College Student

Students use the application to schedule study sessions, track assignment deadlines, and organize coursework. Tasks include uploading notes or projects, creating study plans, and tracking progress on courses. These tasks can occur daily, and some common errors may be missed deadlines or issues scheduling. The user environment includes personal laptops and campus computers. Data items consist of notes, and calendar events.

Use Case 2: Skill Building

Actor: College Student/University Career Center

Students can access training modules to build professional or technical skills relevant to courses or employment opportunities. Tasks include tracking module progress and completing modules. Errors may include incomplete modules or if module progress is not saved correctly. These tasks can occur on a regular basis throughout the semester. Data items consist of module completion status, certifications, and performance metrics.

Use Case 3: Job Preparation

Actor: College Student/Counseling Services

Students use the application to prepare resumes, track internship opportunities and applications, and practice interview skills. Tasks include uploading resumes, setting application deadlines, and accessing forums for recommended interview tips. Some errors that may occur include incorrect file uploads, and missing information on applications or resumes. These tasks are done as needed or during recruitment seasons. The application is accessed from home or campus environments. Data items consist of resume files, internship applications, and interview preparation notes.

Use Case 4: Mental Wellness

Actor: College Student/Academic Programs

Students can use the application to access campus wellness resources, monitor stress levels, and schedule self-care activities. Tasks include scheduling wellness checks and reviewing support services. Errors that may occur are incomplete forms or miscommunication in required wellness resources. These tasks are performed on an as needed basis. Data items include wellness check-in forms and wellness documentation.

3. List of Main Data Items and Entities

1. Guest User

- a. A guest user can browse through and look at public information and content without needing to sign in.
- b. A guest user cannot create a dashboard, save items, or send messages.

2. Registered Student

- a. A registered student can make and manage their personal dashboard, goals, projects, and milestones.
- b. A registered student can send and receive messages within the app.
- c. A registered student can save content/resources and can share them publicly as well.

3. Content Publisher

- a. A content publisher would be a faculty member, an academic program, or a campus support organization.
- b. A content publisher can create, edit, and post resources/content for students to view and discover.
- c. A content publisher cannot access private student dashboards or private messages.

4. Administrator

- a. An administrator manages the settings for the platform and moderates content.
- b. An administrator can remove content, handle reports, and disable accounts that violate platform guidelines.

Core data items and entities

1. User Account

- a. A user account represents a login identity and access level for a registered user.
- b. A user account determines the user type and permissions.

2. User Profile

- a. A user profile stores information provided by the user that is used for personalization, such as interests, challenges, workload, and goals. This information is used to customize the dashboard experience and organize content by relevance.
- b. A user profile also stores user preferences, including privacy choices.

3. Dashboard

- a. A dashboard is the main workspace for a registered student.
- b. A dashboard summarizes progress and provides quick access to goals, projects, reflections, and saved items.

4. Goal

- a. A goal is a user-defined objective that the student wants to achieve.
- b. A goal can have a status, a target date, and optional categories.

5. Project

- a. A project group's related work is structured into a plan with defined outcomes.
- b. A project can contain milestones and can be linked to goals and resources.

6. Milestone

- a. A milestone is a measurable checkpoint within a project.
- b. A milestone helps track incremental progress toward completion.

7. Reflection

- a. A reflection is a student-written entry used to document progress, learning, and challenges over time.
- b. A reflection can be linked to goals, projects, milestones, and resources.

8. Resource

- a. A resource is a discoverable item that helps a student achieve goals, such as a tool, template, workflow, campus service, opportunity, or guide.
- b. A resource includes descriptive metadata such as title, description, URL/link, categories, tags, and optional ratings to support searching and filtering.

9. Saved Item

- a. A saved item is a relationship between a registered student and a resource they want to revisit.
- b. Saved items support personal organization and planning.

10. Message

- a. A message is an in-app communication sent between registered users.
- b. Messages support collaboration, accountability, and peer support.

11. Notification

- a. A notification is an in-app alert that informs a user of important events, such as upcoming due dates or new messages.
- b. Notifications support awareness without requiring an external email.

12. Tag/Category

- a. A category is a classification label used to organize resources and goals for easier discovery.
- b. Categories include use cases such as studying, career prep, and wellness. Skill area: technical, communication, leadership. Content type: tool, template, workflow

13. Search Query

- a. A search query represents a user's request to discover resources or content based on keywords and filters.
- b. Search queries include search terms, selected filters (category, use case, skill area), and sort preferences(relevance, date added, popularity)
- c. Registered students can save frequently used searches for quick access

14. Rating review

- a. A rating is a score (1-5 stars) assigned by a registered student to a resource to indicate its usefulness.
- b. Ratings aggregate to produce an average score that helps other students discover high-quality resources.
- c. Rating influences search result ranking and popularity indicators.

4. Initial list of Functional Requirements

As a fledgling start-up consisting of 6 students, our goal is to develop a **web-based intelligent platform that acts as a guide and framework for college students**. This web application will be designed to help students design, manage, and continuously improve their

lives academically, professionally, and personally. The web application will function as a **life management and discovery system**, integrating **goal setting, intelligent search, reflection, learning analysis, and ethical AI usage guidance** all into one platform. Students will also be able to create a **personal dashboard**, in which they can manage goals, projects, activities, and personal development in areas such as academic success, career preparation, leadership skills, life skills, and overall wellness.

The following are a set of functional requirements that must be adhered to when envisioning the services and functions of the application. These specifications are initial, vague, incomplete, and high-level descriptions that are subject to change.

- 1. A powerful search and discovery engine:** This web application will enable users to find and explore the following:
 - a. Goals and project templates
 - b. Productivity and study workflows
 - c. Academic and career resources
 - d. Improvement tools and learning aids
 - e. Campus-related opportunities and services
 - f. Public community-shared content
- 2. Intelligent search:** This web application will enable users to filter searches with specific keywords and criteria, the following in which includes:
 - a. Category (academic, career, wellness, leadership, finance, creativity, research, etc.)
 - b. Use case (studying, job preparation, mental wellness, skill building, productivity, project management)

- c. Content type (tools, templates, workflows, reflections, resources, events)
 - d. Skill area (technical, soft skills, leadership, research, health, communication)
 - e. Non-AI resources
 - f. Free vs. paid (informational only, no transactions)
- 3. User Account Registration and Login:** This web application will enable users to create an account, log in, and log out to access personalized features like dashboards, saved items, and messaging. This includes:
- a. Basic account registration using a unique email or username
 - b. Secure sign-in and sign-out
 - c. A way for users to reset access if they forget their password
- 4. User Roles and Permissions:** This web application will enforce different access privileges for guest users, college student users, faculty and academic program users, university support organization users, and admins. This includes:
- a. Guest users can browse public content/resources
 - b. College student users can create dashboards, goals, projects, milestones, and save items.
 - c. Faculty and academic program users can publish and update academic resources
 - d. University support organization users can publish and update campus services and opportunities
 - e. Administrators can moderate public content and manage user status
- 5. Personal Dashboard:** This web application will enable college students to create and use a personal dashboard to view and manage goals, projects, milestones, reflections, saved items, and progress summaries. This includes:

- a. A dashboard home view that summarizes what the student is working on right now
 - b. Quick access to goals, projects, and milestones
 - c. An area to look at recent reflections and resources
 - d. A place where students can see a simple overview of progress across their areas, such as academics, career, and wellness
- 6. Goal Management:** This web application will enable college student users to create, view, edit, and delete goals and track goal status over time. This includes:
- a. Short-term and long-term goals
 - b. Goal status such as started, in progress, or completed
 - c. Optional target dates and notes
 - d. Categories like academics, career, wellness, and creativity
- 7. Project Management:** This web application will enable college student users to create and manage projects that organize goals and activities into a structured plan,s such as:
- a. Creating and editing projects that are a part of something bigger, such as a semester plan or internship prep
 - b. Adding goals to a project so that they are not isolated
 - c. Viewing a project overview that shows current milestones and progress
- 8. Milestone management:** This web application will enable college student users to add milestones to projects and update milestone progress over time. This includes:
- a. Creating milestones that act as checkpoints for a project
 - b. Updating milestone status, such as not started, in progress, or completed
 - c. Optional due dates so students can track upcoming checkpoints

- d. Viewing milestones in a simple, ordered list inside each project

9. Reflection and journaling: This web application will enable college student users to write reflections weekly or on demand to document progress, learning, and challenges.

This includes:

- a. Creating, viewing, editing, and saving reflections
- b. Optional reflection prompts that guide the student to write about progress and obstacles
- c. Linking reflections to goals, projects, milestones, and resources so students can connect actions to outcomes
- d. Viewing reflection history so students can review patterns over time

10. Save and organize discovered content: This web application will enable college student users to save resources, templates, workflows, and opportunities for later and organize them inside their dashboard. This includes:

- a. Saving and removing items from a personal saved list
- b. Grouping saved items into simple collections such as study, career, and wellness
- c. Attaching saved items to goals, projects, or milestones so students can use them at the right time

11. Resource publishing for campus groups: This web application will enable faculty and academic program users and university support organization users to publish and maintain resources and opportunities for students to discover through search and browsing. This includes:

- a. Creating and updating resource listings such as campus services, events, programs, guides, and opportunities

- b. Assigning basic categories that match the search filters, such as category, use case, content type, and skill area
- c. Editing and removing outdated resources so the system stays current
- d. Marking resources as free vs paid for informational purposes only

12. In-app messaging and notifications: This web application will enable registered users to send and receive in-app messages and receive in-app notifications for important events. This includes:

- a. Sending and receiving messages between registered users inside the platform
- b. Viewing message threads so conversations are easy to follow
- c. Notifications for events such as new messages, upcoming milestone due dates, and important updates to saved items
- d. A notifications area where users can review recent alerts and clear them when done

13. Personalization and discovery of new content: This web application will recommend new content based on the user's interests, challenges, workloads, and aspirations. The platform may use the user's information and activity to do the following:

- a. Recommend relevant categories, tools, and workflows
- b. Personalize search results
- c. Suggest resources aligned with current goals
- d. Generate insights and growth summaries

14. Interactive Dashboard: This web application, along with user personalization of the dashboard, will offer users access to various tools to enhance their experience as well as interactivity with the community. These features include:

- a. Browsing and searching public goal templates and workflows
- b. Sharing resources, tools, and strategies with other users
- c. Discovering popular or trending student tools
- d. Rating, tagging, bookmarking, and commenting on content

15. Archive for templates: This web application will keep an archive of simple templates for workflow and goals, giving users access to some clean, beginner tools that can be modified to their liking. This will include the following:

- a. Templates and workshops for resume building
- b. Templates and tips on how to organize study guides
- c. Starter templates of various writing formats for any essay

16. Importing of User's personal university resources: This web application will allow users to import certain university information about themselves, giving them quick access to class schedules, GPA, completed courses, and remaining required courses. This includes the following:

- a. Allow users to submit their unofficial transcripts, importing their academic statistics to their dashboard
- b. Give users reminders and notifications about upcoming deadlines, such as class registration
- c. Update users' course schedule and catalog, based on completed courses and any remaining required courses for graduation

17. AI Workshops: This web application will offer workshops on proper and ethical AI usage in the workplace and in school. This does not mean that AI functionality is guaranteed for the final product. Some AI workshops might include:

- a. Video seminars from workers in the industry about proper AI usage in the workplace
- b. Tutorials on fact-checking and error clean-up from AI-generated outputs
- c. Allow users to look up their university's policy on AI usage

18. Accessibility: This web application will be made available and usable to as many people as possible, as well as offering various settings to enhance the user experience. This means that the following features may be included:

- a. Light/Dark mode setting for the user interface
- b. Adjustment of text size across the site
- c. High-contrast color modes for text, images, and backgrounds for easy readability
- d. Font customization
- e. Various color blind adjustment modes

19. Exporting of documents: This web application will allow users to export, print, and/or download various documents based on their work and templates. The following documents are some of the few things that will be made available for exporting:

- a. Resumes
- b. Cover letters
- c. Project diagrams and graphs
- d. Study guides and other notes

20. Document user time and check in on user: The web application will document the amount of time a user has been logged in to the site, as well as how long the user has been active or inactive. These are the following features are included:

- a. Document the total amount of time a user has been logged in for

- b. Document the total amount of time a user has been inactive while logged in
- c. Send notifications to the user's webmail or dashboard if they are inactive for some time
- d. Automatically log out a user if they are inactive for too long

5. Non-functional Requirements

The following are a set of non-functional requirements that must be adhered to when delivering the web application:

1. The application shall be delivered, tested, and deployed using the tools and the server(s) that were approved by the Class CTO and agreed upon during the course of Milestone 0.
2. The application shall be optimized for standard desktop/laptop browsers, and must render correctly on the two (2) latest versions of at least two (2) major browsers.
3. Selected application functions must render well on mobile devices.
4. Data shall be stored in the team's chosen database technology that is located on the deployment server.
5. Privacy of users shall be protected, and all privacy policies shall be clearly communicated to users.
6. English shall be the main language of use.
7. The application shall be very easy and intuitive to use.
8. Google Maps and Google Analytics shall be added wherever appropriate. The following events and resources are deemed to be appropriate:
 - a. Campus resources
 - b. Study locations

- c. Event discovery
 - d. Usage analytics
 - e. etc.
9. No external e-mail clients shall be allowed, and only in-site webmail shall be used.
 10. Pay functionality, if any, shall not be implemented nor simulated in the user interface.
 11. To adhere to and strengthen site security, basic best practices shall be applied, as they are covered in class.
 12. Modern Software Engineering processes and practices shall be used as specified in the class, which includes collaborative and continuous development.
 13. The website shall prominently display the following exact text on all pages at the top of the page:

**“SFSU Software Engineering Project CSC 648-848, Spring 2026. For
Demonstration Only”**
 14. AI-powered assistant(s) and intelligent search enhancements are **optional (bonus requirements)** and must be implemented modularly.

6. Competitive Analysis

To understand LifeTrack's market position, we analyzed popular existing tools that students use for organization and work management. While these platforms are useful, they generally focus on a single area of a student's life, such as task management, scheduling, or note-taking. They do not offer a unified system for managing academic, professional, and personal goals concurrently. Our primary competitors in this space include Notion, Todoist, Google Calendar, and Trello.

Feature / Platform	LifeTrack	Notion	Todoist	Google Calendar	Trello
All-in-one student life management (goals, projects, reflections, resources)	Yes	Partial	No	No	No
Goal, project, and milestone tracking	Yes	Yes	Partial	No	Partial
Resource discovery for student tools and campus services	Yes	No	No	No	No
Reflection and journaling for personal growth	Yes	Partial	No	No	No
Personalization based on student goals and interests	Yes	Partial	Limited	Limited	Limited
Designed specifically for college students	Yes	No	No	No	No

Summary:

Tools such as Notion, Todoist, Google Calendar, and Trello are widely used by students, but they are mostly general-purpose productivity applications. Students often need to combine

multiple tools or spend significant time customizing them to fit their academic and personal needs. LifeTrack is different because it brings goal management, project planning, milestone tracking, reflections, and resource discovery into a single platform that is designed specifically for college students. Instead of switching between several apps, students can use one unified system that focuses on their academic, professional, and personal growth in a more organized and meaningful way.

7. HL System Architecture & Technologies

Overview

Our project, LifeTrack, is a browser-based app for laptop/desktop users that provides a personalized student dashboard in addition to a discovery/search experience for templates, ongoing work, campus resources, and goals. This system is designed to be a web app with separation between the data layer and UI, application layer to provide maintainability and scalability for the future, specifically for search, analytics, and data modeling.

High Level

- Client / Presentation Layer – Web UI rendered within the browser. With support for PC primarily and mobile when/where appropriate.
- Data / Persistence Layer – A MySQL database which stores persistent application data (users, app content, etc.)
- Server / Application Layer – A Node.js + Express backend which handles authentication/session handling, routing, and communication between frontend and backend.
- Optional / Bonus Modular Enhancements – AI assistant and intelligent search features are considered optional, so they need to stay modular.

Constraints for architecture choices

- No payment or simulated pay functionality
- In-site messaging only, no external email
- Needs to render accurately on the two latest versions of at least two major browsers.
- Google Maps & Google Analytics added where appropriate.
- Required Banner text on each page – “SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only.”

Main Software Components

Deployment and Hosting Platform

This application is going to be deployed on a virtual server hosted on Google Compute Engine.

- Server Host: GCE – Google Compute Engine
- Instance Size: e2-micro (1 vCPU, 1GB RAM)
- OS: Ubuntu 22.04 LTS
- Web Server / Reverse Proxy: NGINX 1.22
- SSL / HTTPS: Let's Encrypt using Certbot
- NGINX will be serving as the public entry point and reverse proxy for requests to the Node/Express application
- HTTPS is enforced with the Let's Encrypt Certbot
- The backend app runs continuously using PM2 as a process manager.

Backend Technologies

The server-side app is built using JavaScript and Node.js

- Server-side language: JavaScript (Node.js 20.x LTS)

- Web Framework: Express.js 4.18
- Process Manager: PM2 5.x (keeps server running & process monitoring)
- Package Manager: npm 10.x
- Environment Variables: dotenv 16.x (config for credentials and environment)
- CORS Handling: cors 2.x
- Request Body Parsing: express.json()
- Security Headers: Helmet 7.x

Backend provides API endpoints that the frontend calls for app functionality, and it communicates with MySQL.

Database Technologies

Persistent data, such as user accounts, project data, etc., will be stored in MySQL.

- Database: MySQL 8.0
- Database Driver: mysql2 3.x

Frontend Technologies

Will be built as a standard browser-based UI.

- Frontend Languages: HTML5, CSS3, JavaScript (ES6+)
- UI Framework: Bootstrap 5.3 (used to assist with consistent styling and navigation)

Development and Testing

- IDE: Visual Studio Code
- API Testing: Postman
- Source Control: Git and GitHub (Team repo, workflow, branching)

Supported Browsers and Platforms

- Browsers: Google Chrome and Mozilla Firefox (latest two versions)

- Platforms: Desktop & laptop browsers (primary targets) with support for smaller (mobile) screens)

8. Team and Roles

Team Lead & Github Master:

- Khaterina Sengchareune

Back-end Engineers:

- John Carlo Naranja
- Alex Hernandez

Front End Engineers:

- Lasiru Weerasuriya
- Will Brust

Database Engineer:

- Mohit Kumar

9. Checklist

- Team found a time slot to meet outside of the class: On Track
- Team roles assigned: Done
- Github master chosen: Done
- Team agreed on the listed SW tools and deployment server: Done
- Team ready/learning chosen frameworks: On Track
- The team lead ensured everyone read the final M1:
- Github organized (branches/folders/milestones): On Track